



Frequency-Domain Enhanced Extreme Bandwidth Extension Network with ICCRN for Superior Speech Quality

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Abstract

Bandwidth extension is a critical task in speech processing that enhances narrowband audio quality by reconstructing essential high-frequency components, which are vital for natural and intelligible speech. Recent advancements like EBEN (Extreme Bandwidth Extension Network) achieve notable results with lightweight generators and real-time operations. However, down-sampling limitations degrade frequency-space information, causing speech distortion as up-sampling struggles to recover lost details. To address this, we propose replacing the generator with an ICCRN in the frequency domain. Leveraging cepstral-domain features and in-place convolutions, our approach mitigates information loss during down-sampling and preserves global spectral structure. Evaluations on the French LibriSpeech dataset show significant improvements, including clearer spectrograms, fewer generator parameters, and lower computational costs, advancing the state of the art in bandwidth extension.

Index Terms: bandwidth extension, frequency-space information, ICCRN, cepstral-domain features, in-place convolutions

1. Introduction

In recent years, deep learning techniques have been incorporated into bandwidth extension methods, significantly enhancing the audio quality and naturalness of bandwidth extension systems. Time-domain bandwidth extension methods[1, 2, 3, 4, 5] utilize neural networks to map narrowband speech waveforms and directly generate or predict wideband speech waveforms. Frequency-domain bandwidth extension techniques[6, 7, 8, 9, 10] typically employ spectral features from the lower bands (such as complex spectra, magnitude spectra, etc.) as inputs for neural models, with the learning target generally being the clean speech spectrum in the high-frequency bands or masks used to estimate the high-frequency parts (for example, ideal ratio masks, complex ideal ratio masks, etc.).

The goal of bandwidth extension technology is to synthesize or estimate missing high-frequency components by using low-frequency information, thereby improving the quality and perceptual effect of speech or audio signals. This approach is particularly suitable for telephone communications or other scenarios where high-frequency loss occurs due to bandwidth limitations. Through deep learning models, patterns related to high-frequency content can be learned from low-frequency data, and reasonable high-frequency reconstruction can be performed based on this, making the restored audio sound more complete and natural.

EBEN[3] leverages a multiband decomposition of the raw captured speech to decrease the data time-domain dimensions, and gives better control over the full-band signal. This multiband representation is fed to a U-Net-like model, which adopts a combination of feature and adversarial losses to recover an corrupted audio signal. This method achieves excellent performance on the French LibriSpeech dataset[11] with a lightweight generator and real-time compatible operation.

Despite its excellent performance, EBEN still has certain limitations. Firstly, compared to frequency-domain processing, time-domain methods may be limited by the constraints of time series analysis, making it challenging to directly capture changes in the spectrum. In contrast, frequency-domain approaches provide a clear representation of a signal's spectral structure, enabling more precise control over the bandwidth extension process. By manipulating frequency components, frequency-domain methods can achieve more flexible and accurate bandwidth extension. Moreover, the downsampling operation in EBEN might disrupt the spatial information of frequencies, and the upsampling process cannot fully restore this information, which could lead to a deeper level of speech distortion.

This study further improves EBEN to enhance bandwidth extension performance. Firstly, the method is modified to perform bandwidth extension in the frequency domain, allowing for more precise control over the extension process. The multiband decomposition of the raw captured speech is retained, but with a difference: the decomposed bands are transformed into the frequency domain via Short-Time Fourier Transform (STFT) for frequency-domain operations. Secondly, considering that the original generator's downsampling might cause a deeper level of speech distortion, the generator is replaced with ICCRN[12] (Inplace Cepstral Convolutional Recurrent Network), which effectively utilizes cepstral domain features and in-place convolution to address the issues of downsampling while preserving global information. Experimental results demonstrate that the improved EBEN model not only has fewer parameters and lower computational requirements compared to the original EBEN but also exhibits superior performance.

2. Method

2.1. Pseudo Quadrature Mirror Filter formalism

In this work, we also employ the Pseudo Quadrature Mirror Filter (PQMF) banks implemented and provided in EBEN for audio processing. This technique was initially proposed in[13]. The QMF banks utilize a set of analysis filters $\{H_i\}_{i \in [1, M]}$ to decompose a signal into multiple non-overlapping subband channels with equal bandwidth, and synthesis filters $\{G_i\}_{i \in [1, M]}$ for subsequent signal reconstruction. These filters are derived from

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the same lowpass prototype filter. A typical frequency response for an M-band PQMF bank is shown in Fig. 1, and the entire processing pipeline is illustrated in Fig. 2.

Theoretically, exact reconstruction can be achieved if the analysis filters $\{H_i\}_{i \in [1, M]}$ and synthesis filters $\{G_i\}_{i \in [1, M]}$ have infinite support.

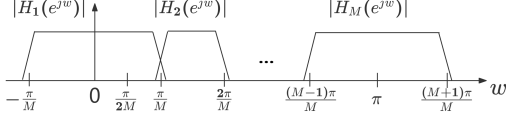


Figure 1: Frequency response of a PQMF filter bank

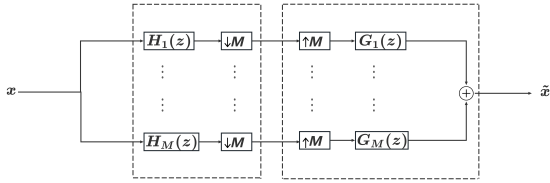


Figure 2: PQMF Analysis and Synthesis : block-diagram

In subband coding, PQMF analysis filter outputs are downsampled by a factor of M. This accelerates inference and allows different frequency bands to be modeled as conditionally independent. PQMF analysis thus reduces channel redundancy and computational complexity. In our experiments, we used EBEN’s efficient PQMF implementation with native PyTorch functions, relying on strided convolutions for high performance and compatibility.

2.2. ICCRN Model

Figure 3 depicts ICCRN’s architecture. As an inplace model, it eliminates frequency downsampling or upsampling, reducing information loss and avoiding the introduction of irrelevant data. The frequency dimension stays fixed at $f=161$, with all convolutional layers having $c=20$ output channels. ICCRN uses cepstrum analysis to examine time-frequency (T-F) domain features. A real-valued FFT is applied to each channel’s T-F feature map per frame. In the resulting cepstral space, the spectral envelope energy, containing most semantic information and timbre cues, concentrates in the low cepstral region, while harmonics appear as sparse peaks distributed periodically across higher cepstral bands[14]. By analyzing harmonic structures, ICCRN predicts and synthesizes high-frequency components beyond the original bandwidth, achieving high-quality extension that maintains natural audio characteristics. This harmonic-based method improves audio quality and preserves timbral fidelity, ensuring accurate signal representation.

We utilize a bidirectional LSTM operating channel-wise (F-chBLSTM) to transform the input channel dimension into a higher-dimensional representation. Once this transformation is complete, the resulting features are processed by n sequential CFBs. Next, a two-layer channel-wise LSTM (T-chLSTMx2) creates a mask for the encoder’s output, with its hidden layer size adjusted to $2c$. During decoding, the masked feature map, enriched by skip connections, is processed through n stacked CFBs and further refined by a time dimension channel-wise LSTM. The channel dimension is then expanded to $2 \times m$ channels using a 1×1 convolutional layer, where m corresponds to

the number of bands produced by the PQMF banks decomposition. This channel arrangement models the concatenation of complex spectra across different frequency bands, obtaining the predicted high-frequency components, with the factor of 2 representing the real and imaginary parts of the estimated STFT spectrum. Ultimately, by combining the low-frequency signal from the input and the predicted high-frequency components in the time-frequency domain, we synthesize an extended bandwidth audio signal. This process effectively enhances the frequency range of the original narrowband signal by generating accurate high-frequency components that complement the existing lower frequencies, thereby significantly improving audio quality.

2.3. Details of the method

First, the 16kHz sampled speech signal is decomposed into M equal-bandwidth, non-overlapping sub-bands in the time domain using PQMF banks. Among these, the lowest-frequency first sub-band undergoes a Short-Time Fourier Transform (STFT), converting it into a dual-channel complex spectrum, which is then fed into the generator ICCRN. During the encoder-decoder process, ICCRN progressively generates mid-to-high frequency information, ultimately outputting an eight-channel complex spectrum. Each pair of these eight channels represents the real and imaginary parts of the complex spectra for one of the four sub-bands. By processing only the low-frequency information contained in the first sub-band, this method significantly reduces information redundancy and greatly decreases computational complexity. After network processing, the complex spectra of the four sub-bands are subsequently transformed back to the time domain via inverse Short-Time Fourier Transform (iSTFT), thereby recovering the four time-domain sub-bands. Finally, PQMF banks is used once again to reconstruct the original audio signal from these four sub-bands, restoring its full spectral range. The discriminator design draws inspiration from the architecture of the discriminators in EBEN. It employs multiple discriminators to directly process the sub-band signals decomposed by PQMF banks, without the need to recombine or upsample these sub-band signals. Unlike EBEN, we input not only the generated high-frequency bands but also the original low-frequency band. This approach allows the discriminator to obtain more contextual information, aiding it in more accurately evaluating whether the generated high-frequency bands are reasonable and natural. To accommodate this multi-band input, we have adjusted the parameters of each layer in the discriminator accordingly. Additionally, to ensure consistency across different frequency bands, we use a full scale MelGAN discriminator identical to the one in EBEN. This further validates the quality and overall consistency of the generated audio.

2.4. Loss functions

The discriminators are trained using a classical hinge loss. Within this framework, G denotes the generator, and $D_{k,t}^{(l)}$ represents layer l (among L_k layers) of the discriminator at scale k (among K scales) and time t . The notations $F_{k,l}$ and $T_{k,l}$ refer to the number of features and the temporal length, respectively, for given indices. Here, x signifies the in-ear signal, whereas y stands for the reference signal.

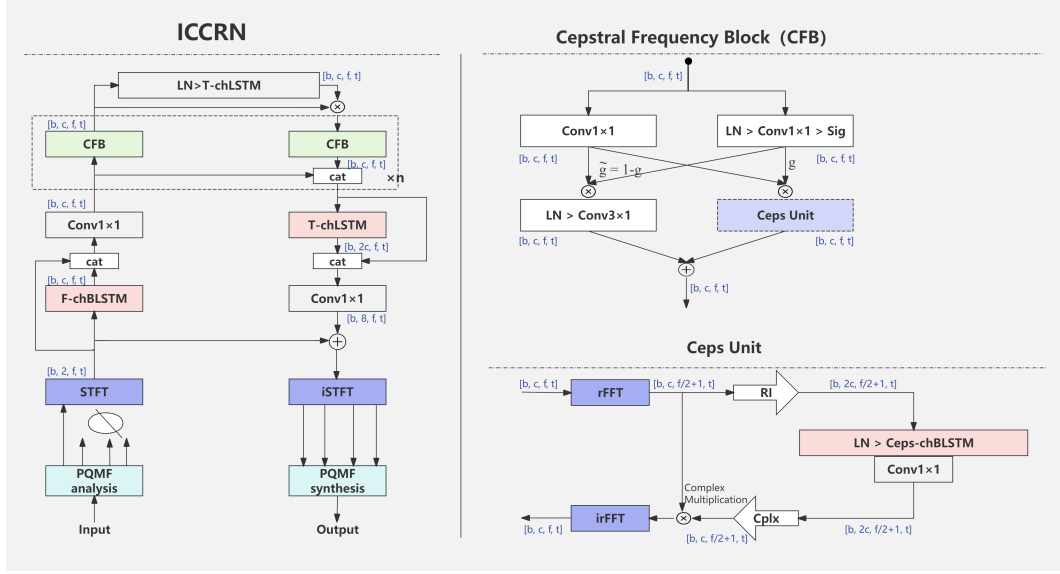


Figure 3: The implemented ICCRN architecture.

$$\begin{aligned} \mathcal{L}_{\mathcal{D}} = & E_y \left[\frac{1}{K} \sum_{k \in [0,3]} \frac{1}{T_{k,L_k}} \sum_t \max(0, 1 - D_{k,t}(y)) \right] \\ & + E_x \left[\frac{1}{K} \sum_{k \in [0,3]} \frac{1}{T_{k,L_k}} \sum_t \max(0, 1 + D_{k,t}(G(x))) \right] \end{aligned} \quad (1)$$

Similarly, for the generator, we utilized a loss function that consists of a generative component $\mathcal{L}_{\mathcal{G}}^{adv}$ and a reconstructive component $\mathcal{L}_{\mathcal{G}}^{rec}$, balanced by a weighting factor λ . The reconstruction loss corresponds to the "feature" loss introduced in[15], which is essentially the mean absolute error between the internal layer activations of the discriminator for the synthesized audio and those for the matching reference audio. The overall generator loss is given by:

$$\mathcal{L}_{\mathcal{G}} = \mathcal{L}_{\mathcal{G}}^{adv} + \lambda \mathcal{L}_{\mathcal{G}}^{rec}. \quad (2)$$

$$\mathcal{L}_{\mathcal{G}}^{adv} = E_x \left[\frac{1}{K} \sum_{k \in [0,3]} \frac{1}{T_{k,L_k}} \sum_t \max(0, 1 - D_{k,t}(G(x))) \right] \quad (3)$$

$$\mathcal{L}_{\mathcal{G}}^{rec} = E_x \left[\frac{1}{K} \sum_{\substack{k \in [0,3] \\ l \in [1, L_k]}} \frac{1}{T_{k,l} F_{k,l}} \sum_t \|D_{k,t}^{(l)}(y) - D_{k,t}^{(l)}(G(x))\|_{L_1} \right] \quad (4)$$

Through this combination, the generator ensures high fidelity to the reference signal while exhibiting creativity in the high-frequency regions, particularly where information is scarce in the degraded signal (such as for fricatives). Specifically, the reconstruction loss $\mathcal{L}_{\mathcal{G}}^{rec}$ guarantees that the generated audio samples closely match the original reference, whereas the adversarial loss $\mathcal{L}_{\mathcal{G}}^{adv}$ introduces flexibility and innovation in the higher frequencies. Importantly, the reconstruction loss operates not directly in the time domain but within the discriminator domain, focusing on semantic feature matching of the signal.

3. Experimental Setup

3.1. Datasets

We evaluated our model on the French LibriSpeech dataset[11], using the same processing and degradation settings as EBEN. Specifically, the degradation uses a 2^{nd} order low-pass filter (600 Hz cutoff, unitary Q-factor) applied via *filtfilt* to ensure zero phase shift, and adds Gaussian white noise 23 dB below the signal power.

3.2. Configuration

for STFT, we adopt 320/80 for window-length/hop-length, and the analysis uses a Hamming window. We use 320-point Fast Fourier transform (FFT) to extract 161-dimensional complex spectra for 16 kHz sampling rate. The model is optimized by Adam. The learning rate is 0.0003. The batch size is 16. The parameter M for the PQMF banks is set to 4. The generator's network architecture aligns with that of ICCRN, utilizing 20 channels and employing a discriminator similar to the one used in EBEN. To verify the effectiveness of the proposed method, we selected EBEN as a baseline model and conducted comparisons using the same evaluation protocol as EBEN. Additionally, we compared our approach with other methods, including Audio U-net[16], Hifi-GAN v3[17], Seanet[18], Strm-Seanet[5].

4. Experiment Results and Analysis

In this section, We compare our method with other baseline systems across various metrics, including SI-SDR[19], STOI[20], PESQ[21] and Parameters(M), and show the results on the French Librispeech test dataset. SI-SDR is Scale-invariant Signal-to-Distortion Ratio. STOI is Short-Term Objective Intelligibility. PESQ is Perceptual Evaluation of Speech Quality. These three metrics are all better when higher. "#Para" indicates the number of parameters for both the discriminator and generator, with the unit being millions (M).

In Table 1, Ours v1 refers to the method after replacing the generator, and Ours v2 further modifies the discriminator. Com-

Table 1: Comparing models with: PESQ/SI-SDR/STOI on test set — MUSHRA-U (88 participants) and MUSHRA-Q (82 participants) scores — number of parameters. Format is median (IQR). MUSHRA-U is about ease of understanding while the MUSHRA-Q is about audio quality.

Model	Speech Metrics					#Para(M)	
	PESQ	SI-SDR	STOI	MUSHRA-U	MUSHRA-Q	Gen	Dis
Simulated In-ear	2.42 (0.34)	8.4 (3.7)	0.83 (0.05)	51 (29)	24 (18)	—	—
Audio U-net[16]	2.24 (0.49)	11.9 (3.7)	0.87 (0.04)	60 (26)	33 (18)	71.0	—
Hifi-GAN v3[17]	1.32 (0.16)	-25.1 (11.4)	0.78 (0.04)	40 (23)	36 (18)	1.5	70.7
Seanet[18]	1.92 (0.48)	11.1 (3.0)	0.89 (0.04)	73 (13)	78 (12)	8.3	56.6
Strm-Seanet[5]	2.01 (0.46)	11.2 (3.6)	0.89 (0.04)	66 (20)	61 (14)	0.7	56.6
EBEN[3]	2.08 (0.45)	10.9 (3.3)	0.89 (0.04)	73 (14)	76 (14)	1.9	27.8
Ours V1	2.16 (0.53)	11.3 (3.3)	0.89 (0.03)	—	—	0.46	27.8
Ours V2	2.34 (0.59)	11.8 (3.6)	0.90 (0.03)	—	—	0.46	28.3

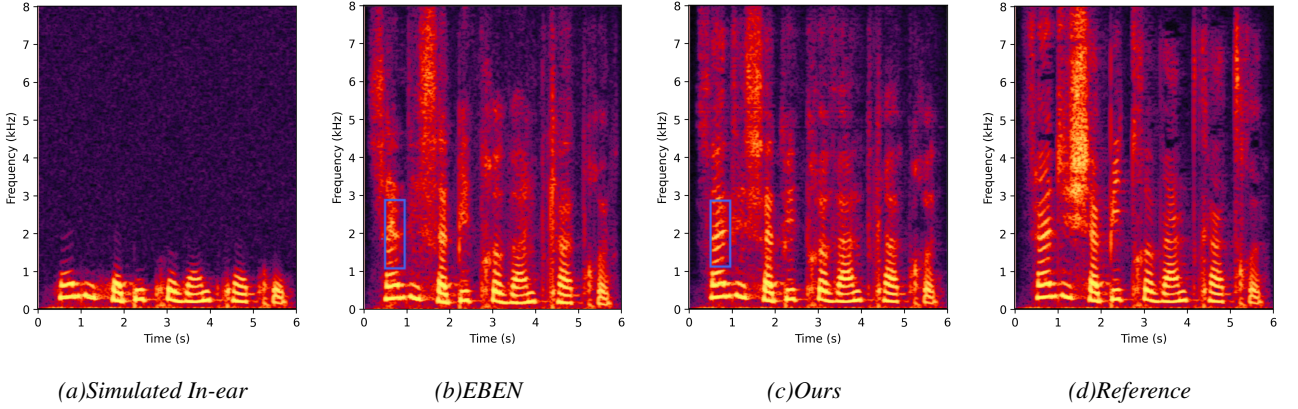


Figure 4: Spectrogram Comparison between EBEN and Ours

Table 2: MUSHRA Comparison between EBEN and Ours (40 Participants). Format is median (IQR).

Method	MUSHRA
EBEN	74.2 (4)
Ours	75.4 (5.4)

paring Ours v1 with EBEN shows the effectiveness of our generator design. Further comparison between Ours v1 and Ours v2 validates the effectiveness of the modifications to the discriminator’s input and layer parameters. Our method outperforms EBEN in all objective metrics while using only about one-quarter of the generator parameters. This makes ours more suitable for resource-constrained environments like mobile devices and embedded systems. As a causal system, our method ensures efficient inference, enhancing practical flexibility.

In Figure 4, we show the input speech, reference speech, and the speech enhanced by EBEN and our method. Our method produces richer high-frequency details and, due to frequency-domain processing, has a clearer harmonic structure, improving spectral complexity and clarity. This method also enhances the generation of harmonic components, leading to better overall audio quality.

In Table 2, we conducted a MUSHRA test specifically for EBEN and our proposed method. The experiments in EBEN compared it with other methods listed in Table 1. For convenience, we directly compared the subjective listening performance between EBEN and our method. The results indicate

that our method performs comparably to or slightly better than EBEN in terms of subjective listening quality. Audio demo available at: <https://github.com/bb1885566/audio-demo>.

We think that using ICCRN as the generator is crucial. It effectively preserves frequency-space information during downsampling, significantly reducing speech distortion and improving objective audio metrics. Additionally, by introducing PQMF banks, we utilize multi-band information for GAN training while maintaining a low parameter count. This frequency-domain approach achieves flexible and precise bandwidth extension, further enhancing model performance.

5. Conclusions

This paper presents an improved bandwidth extension network to address the potential loss of frequency-space information and increased speech distortion during the downsampling process in the original EBEN model. To this end, we replaced the original generator with a new one called ICCRN (Inplace Cepstral Convolutional Recurrent Network) and adjusted the parameters of each layer in the discriminator. The ICCRN effectively mitigates downsampling issues by utilizing cepstral domain features and inplace convolution techniques, preserving global information integrity. Additionally, our method retains the PQMF method for efficient conversion of subband speech signals into the frequency domain. Experimental results show that our method outperforms the original EBEN model in objective metrics and either surpasses or performs comparably in subjective evaluations. Notably, our method generates finer mid-to-high frequency details, resulting in superior audio quality.

6. References

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